The engineering problem

This project aims to develop a machine-learning-based system for localising apples within images of apple trees. Automated apple localisation could support orchard monitoring by reducing the need for manual fruit identification and enabling more consistent apple counting.

The main engineering challenge is distinguishing apples from visually similar objects in complex orchard environments. Leaves, branches, background vegetation and varying illumination can produce false-positive detections, while overlapping fruit and partial occlusion can cause apples to be missed. Detecting apples throughout the entire image may therefore introduce unnecessary detections in regions that are unlikely to contain the target fruit.

The proposed approach uses the characteristic Y-shaped scaffold branches as a structural reference. A deep-learning model predicts the branch mask, which defines an expanded region of interest for apple detection. This region must balance excluding unrelated background objects against retaining apples growing on secondary branches.

The system aims to estimate individual apple locations and counts in two-dimensional images. Its performance will be evaluated against an RF-DETR instance-segmentation [1] reference using detection precision, recall and counting error, although, it should be kept in mind that this segmentation results may not be 100% accurate

Inputs, Outputs and Data

Branch Detection

Inputs

The branch-segmentation system uses a dataset of RGB images, depth images and ground-truth (GT) branch masks collected from apple trees during summer and winter observations.

RGB images

The RGB images are stored as PNG files with dimensions of 640×480 pixels, using 8-bit unsigned integer (uint8) encoding with three colour channels. They provide visual information about the tree structure, including the Y-shaped trunk and primary scaffold branches, as well as the appearance of apples and surrounding foliage. The RGB images serve as inputs to all three branch-segmentation models.

Depth images

The depth images are stored as PNG files with dimensions of 640×480 pixels. Each depth image is converted to grayscale and combined with the three RGB channels to form a four-channel input for branch segmentation. Depth information provides additional scene information that may help distinguish the tree structure from its surroundings.

Ground-truth branch masks

The summer dataset contains 562 GT branch-mask files, stored as 256×256-pixel PNG images with uint8 values ranging from 0 to 255. These masks identify the Y-shaped trunk and primary scaffold branches and provide pixel-level supervision for training and evaluating the three segmentation models. The masks are converted into binary representations using an intensity threshold of 127.

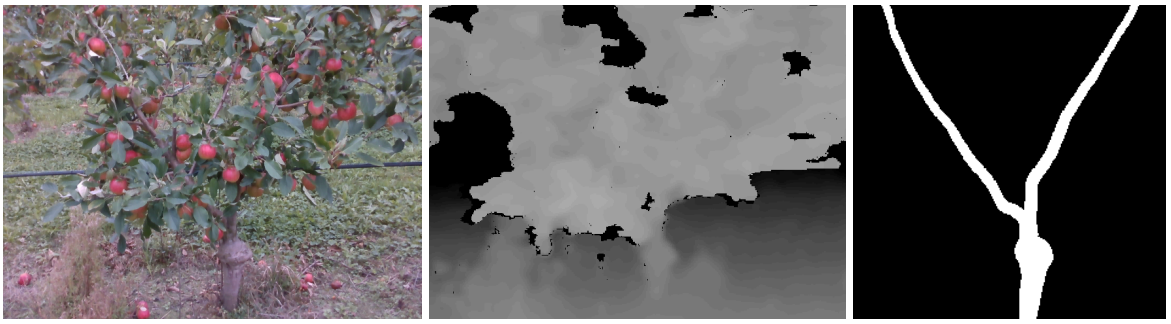


Figure 1: Example Summer Apple Tree Images: RGB Image (Left), Depth Image (Middle), Branch Mask Ground Truth (Right)

Seasonal dataset composition

The dataset contains 562 summer RGB-D observations with corresponding GT branch masks and 562 winter RGB-D observations without GT masks, totalling 2,810 image files. Summer images are used for supervised training and validation, while winter images support qualitative assessment of seasonal generalisation.



Figure 2: Example Winter Apple Tree Images: RGB Image (Left), Depth Image (Right)

Outputs

The system produces outputs from both the branch-segmentation and the apple-detection task. For branch segmentation, each model generates a predicted binary mask at a resolution of 256 x 256 pixels, matching the original ground truth branch masks and identifying the Y-shaped trunk and primary scaffold branches. An example of the predicted branch mask next to its ground truth can be seen in Figure 3 below.

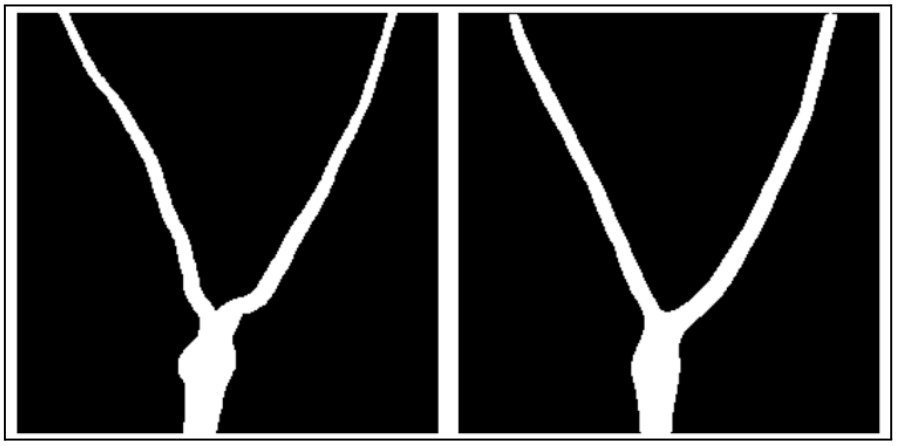


Figure 3: Example Ground Truth Branch Mask (left) next to Predicted Branch Mask (right)

Apple Localisation

Inputs

The apple-localisation system uses the previously described RGB-D images and Y-shaped branch masks. The branch masks define the region of interest, while RGB and depth data support candidate detection and valid-depth filtering.

Validation data

For comparison and validation, an RF-DETR instance-segmentation model [1] was trained using an annotated apple dataset obtained from Roboflow [2]. The trained model was subsequently applied to the 562 summer orchard images to generate apple instance masks, bounding boxes and confidence scores. These predictions served as pseudo-ground-truth references for evaluating the traditional apple detector, rather than independently verified annotations of the orchard dataset.

Outputs

The apple-localisation pipeline outputs two-dimensional bounding boxes and centroids for individual candidate apples within a dilated region surrounding the ground-truth Y-shaped scaffold branches. Detections are numbered and annotated on the original RGB images, with per-image counts recorded in `apple_counts.csv` and individual detection coordinates, areas and branch-overlap values stored in `apple_instances.csv`.

Both the traditional detector and RF-DETR produce comparable instance-level outputs, enabling evaluation of counting differences and spatial agreement. RF-DETR additionally exports instance-segmentation masks and confidence scores, while the traditional detector records a red-dominance score. Depth information is used only to reject invalid pixels in the traditional pipeline.

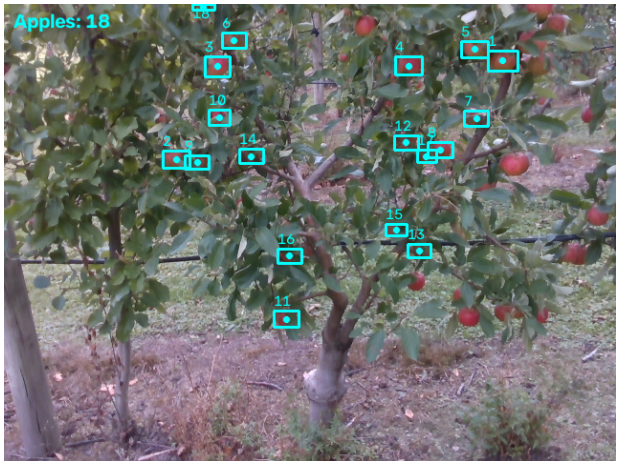


Figure 4: Example Outputs of the apple localiser

The Computational Approach

The project investigates two complementary approaches to apple-tree analysis: deep-learning-based branch segmentation and traditional image-processing-based apple detection. Branch segmentation identifies the trunk and primary scaffold branches, while apple detection identifies potential fruit using colour, depth validity and branch-region information.

1. Branch Detection Using Deep Learning

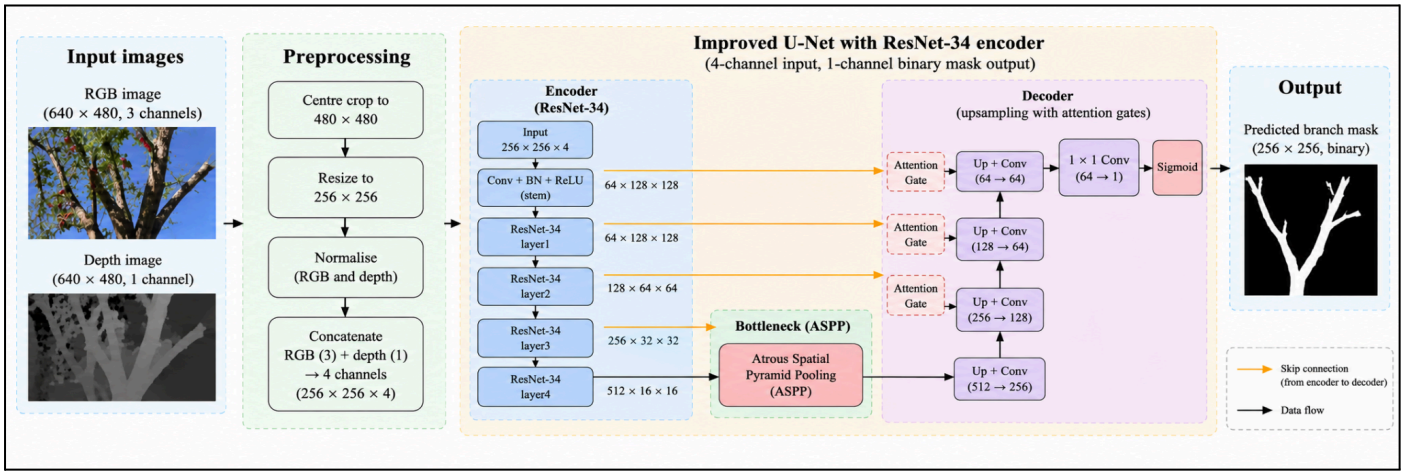


Figure 5: A flowchart showing the computational architecture used

The branch-detection pipeline uses an improved U-Net to perform binary semantic segmentation of the trunk and primary scaffold branches. Its output is a pixel-level mask representing the tree's main structural branches for subsequent analysis.

Model architecture and processing pipeline

The model extends a U-Net encoder–decoder [3] with a pretrained ResNet-34 encoder [4] and several architectural enhancements:

- **Pretrained ResNet-34 encoder [4]:** Extracts hierarchical features from four input channels: RGB and grayscale depth. The first convolution is modified to accept four

channels, retaining pretrained RGB weights and initialising the depth-channel weights using the mean of the original RGB filters.
- **Atrous Spatial Pyramid Pooling (ASPP) [5], [7]:** Captures multi-scale context to support recognition of branches with different widths and appearances.
- **Attention-gated skip connections [6]:** Select relevant encoder features while suppressing background information, helping distinguish branches from leaves, fruit and other structures.
- **Residual decoder blocks [4]:** Refine reconstructed features and support gradient propagation, aiding the recovery of branch boundaries and spatial details.
- **Binary mask generation:** A single-channel output is converted from logits to pixel-wise probabilities using a sigmoid function, then thresholded to produce the final binary mask.

Data preprocessing

The model uses paired summer RGB images, depth images and ground-truth (GT) branch masks. Preprocessing follows the working U-Net pipeline:

1. Centre-crop RGB and depth images to 480 x 480 pixels.
2. Resize RGB images to 256 x 256 using bilinear interpolation and depth images using nearest-neighbour interpolation.
3. Normalise RGB channels using ImageNet means and standard deviations, and scale grayscale depth values to (0,1).
4. Convert GT masks to binary using an intensity threshold of 127, retaining their original 256 x 256 dimensions.

The dataset is divided into training and validation subsets using an 80:20 split. Training-only augmentation includes paired horizontal and vertical flips and mild RGB brightness and contrast variation. Validation samples remain unaugmented.

Training and evaluation

The improved model uses AdamW [8] optimisation, learning-rate scheduling and early stopping based on validation performance. Binary cross-entropy with logits is used as the loss function.

Performance is assessed using four metrics:

- **Dice coefficient:** Measures overlap between predicted and GT branch masks.
- **Intersection over Union (IoU):** Measures the intersection relative to the union of the predicted and GT foreground regions.
- **Precision:** Measures the proportion of predicted branch pixels that are true positives.
- **Recall:** Measures the proportion of GT branch pixels successfully detected.

The checkpoint with the highest validation Dice is retained. Visualisations compare the RGB image, depth image, GT mask, predicted mask and an overlay highlighting correctly and incorrectly segmented regions.

The improved U-Net combines pretrained features, multi-scale context, attention-based feature selection and residual decoding. Its effectiveness is compared with the original U-Net and a basic multilayer perceptron (MLP) baseline using the same data split and evaluation metrics. Any improvement is determined experimentally rather than assumed from the architectural changes.

2. Apple Localisation Using Traditional Image Processing

The apple-localisation pipeline uses traditional image-processing techniques to identify candidate apples near the Y-shaped scaffold branches. Unlike the deep-learning-based branch-segmentation models, this approach applies explicit colour, geometric and spatial constraints, providing an interpretable and computationally inexpensive method for identifying potential fruit. The pipeline consists of colour segmentation, morphological filtering, connected-component analysis, geometric filtering and branch-region validation.

Colour and Depth-Based Segmentation

Candidate apple pixels are initially identified using RGB colour thresholding. A pixel is classified as a potential apple when its red-channel intensity exceeds 50 and is at least 1.5 times greater than both its green and blue channels. These thresholds aim to distinguish red apples from surrounding green foliage and brown branches.

Depth information is incorporated through a validity constraint, which excludes pixels with depth values less than or equal to zero. This prevents invalid depth regions from contributing to detections but does not provide metric three-dimensional localisation.

Morphological Processing and Candidate Extraction

The resulting binary colour mask undergoes morphological opening and closing using a 5 × 5 elliptical structuring element [9]. Opening removes small isolated regions, while closing fills minor gaps within candidate objects. Eight-connected component analysis then groups adjacent foreground pixels into individual apple candidates.

Each component is filtered using its pixel area, bounding-box aspect ratio and circularity. The implementation retains components with areas between 30 and 5,000 pixels, aspect ratios no greater than 2.5 and circularity values of at least 0.20. These relatively permissive geometric constraints accommodate partially occluded or irregularly shaped apples while rejecting small noise regions and highly elongated objects.

Y-Shaped Branch-Region Filtering

To restrict detections to the relevant tree structure, the ground-truth Y-shaped scaffold branch mask is expanded using morphological dilation with a 100 × 100 elliptical structuring element. This creates a spatial region of interest extending beyond the main branches to account for apples growing on nearby secondary branches.

Each apple candidate is evaluated according to the proportion of its pixels overlapping the expanded branch region. Candidates with less than 5% overlap are rejected, reducing detections in unrelated image regions. The dilation size and minimum overlap threshold represent important parameters because a larger search region may retain more apples but introduce additional false positives, whereas a smaller region may exclude valid fruit.

3. Computational Justification and Design Considerations

Branch Detection Using Deep Learning

Deep learning was selected for branch segmentation because varying branch thickness, foliage occlusion and complex backgrounds make fixed image-processing thresholds unreliable. U-Net was chosen for its ability to preserve spatial detail through encoder–decoder skip connections, while pretrained ResNet-34 features reduce reliance on learning representations entirely from the limited labelled dataset. ASPP, attention gates and residual decoder blocks were introduced to address multi-scale structure and background interference, with their combined effectiveness assessed against simpler architectures.

Apple Localisation Using Traditional Image Processing

Traditional image processing was selected for its interpretability, low computational complexity and ability to identify red apple candidates without additional model training. Combining colour and geometric filtering with a dilated branch region allows detections to be restricted to relevant tree structures. The dilation size and overlap threshold provide adjustable trade-offs between false positives and missed apples.

Validation, Evaluation and Limitations

1. Branch Segmentation

The three branch-segmentation models were evaluated using the same 20% summer validation subset, with Dice coefficient, IoU, precision and recall measuring agreement with ground-truth Y-shaped branch masks. The best checkpoint is selected using validation Dice, with predictions and comparison images saved for visual inspection. Winter images without ground-truth annotations were additionally used for qualitative visual comparisons.

However, the validation subset was also used for model selection, and no independent test set was generated to preserve more training images. Generalisation remains unverified across different apple-tree varieties, orchard environments, lighting conditions, camera viewpoints, imaging distances, resolutions and depth-sensor configurations. Winter performance also cannot be quantitatively verified without corresponding annotations. Furthermore, the models specifically target dominant Y-shaped scaffold branches, and their performance on alternative pruning structures or other tree species has not been established.

2. Apple Localisation

The traditional apple detector was evaluated against branch-guided RF-DETR instance-segmentation predictions across 562 summer images. Individual detections were compared using one-to-one centroid matching [10] within a 30-pixel threshold. Counting MAE and RMSE measured differences in estimated apple counts, while precision, recall, F1-score, centroid distance and bounding-box IoU quantified detection and localisation agreement.

RF-DETR predictions served as pseudo-ground-truth references rather than verified annotations. Therefore, the metrics represent agreement between methods rather than absolute detection accuracy. The traditional detector was also evaluated using both

ground-truth and U-Net-predicted branch masks to assess the influence of branch segmentation on downstream apple localisation. However, differences in mask alignment may affect the comparison, and the predicted-branch experiment does not establish independent end-to-end generalisation. Performance remains sensitive to fruit colour, illumination, occlusion and geometric thresholds, while depth information is used only for validity filtering rather than three-dimensional localisation.

Results, Comparisons and Optimisations

Branch Detection

Three architectures were compared to evaluate the effectiveness of the proposed improved U-Net. It should be noted that for neural network training, all training was done on a NVIDIA GeForce RTX 5060.

The **pixel-wise MLP baseline,** which has an architecture depicted in Figure A1 in the appendix, uses fully connected layers to classify each pixel independently based on its RGB-D values. Without spatial feature extraction, it provides a simple computational baseline for assessing the importance of spatial context in branch segmentation.

The **baseline ResNet-34 U-Net**, which has an architecture depicted in Figure A2 in the appendix, employs a pretrained ResNet-34 encoder and a conventional decoder with skip connections. Unlike the MLP, it captures spatial and contextual information through convolutional feature extraction, providing a stronger reference architecture.

The **improved U-Net** previously mentioned in the Computational Approach extends the baseline with ASPP, attention-gated skip connections and residual decoder blocks to enhance multi-scale feature extraction and branch segmentation. All three models were evaluated using the same dataset split and metrics, with their quantitative performance compared in Table 1.

All 3 models were tested on the winter images and were used for visual inspection of the generalisation of the mode. It can be seen from Figure 6, that the improved U-Net generalises well to unseen data of Y-shaped branches



Figure 6: Example Outputs of the 3 Models on the Winter Dataset, Improved U-net (Left), Basic U-net (Middle), Baseline MLP (Right)

Table 1. Comparisons of All 3 Branch Predictors

Model	Training time (min)	Inference time (ms/image)	Validation Dice	Validation IoU
Pixel-wise MLP	9.5	2.198	0.1861	0.1026
Basic ResNet-34 U-Net	8.5	5.103	0.3614	0.2205
Improved U-Net	16.2	7.583	0.5563	0.3853

The results highlight a trade-off between computational efficiency and segmentation accuracy. The pixel-wise MLP offers the fastest inference (2.198 ms/image) but achieves the lowest Dice (0.1861). The basic U-Net improves Dice to 0.3614 with a modest increase in inference time (5.103 ms/image). The improved U-Net achieves the best Dice (0.5563) and IoU (0.3853), but requires the longest training time (16.2 minutes) and inference time (7.583 ms/image). Therefore, the improved U-Net is preferable when segmentation accuracy is prioritised, while the MLP is more suitable when inference speed is critical. Further optimisation should aim to reduce the improved U-Net's computational cost without sacrificing accuracy.

Apple localisation

The traditional image-processing detector was evaluated against branch-guided RF-DETR instance segmentation across 562 summer orchard images. RF-DETR detections were treated as pseudo-ground-truth references, with apples matched through one-to-one centroid assignment using a maximum distance of 30 pixels. Detection agreement, counting error and localisation accuracy were assessed.

Table 2. Apple detection and localisation comparison using RF-DETR as a pseudo-ground-truth reference.

Apple Count (Trad/RF-DETR)		MAE	RMSE	Precision (%)	Recall (%)	F1-score (%)	Centroid Error (px)	BBox IoU (%)
14.94	32.55	17.68	20.06	85.63	39.30	53.88	4.50	40.26

The traditional detector achieved high precision (85.63%) but relatively low recall (39.30%), indicating that most accepted detections agreed with RF-DETR while many reference instances remained undetected. This resulted in substantial undercounting, with a mean absolute error of 17.68 apples per image. The mean centroid error of 4.50 pixels and bounding-box IoU of 40.26% suggest stronger agreement in apple centres than object extents.

A branch-mask sensitivity experiment was also done to compare predicted and ground-truth branch regions across the same 562 images. Using predicted branches increased precision from 85.63% to 96.33%, recall from 39.30% to 43.97%, and F1-score from 53.88% to 60.38%, while counting MAE remained similar (17.71 versus 17.68). This demonstrates that branch-mask selection influences detection agreement without substantially affecting counting error.

To improve detection performance, a grid search evaluated 5,184 parameter configurations across 444 development images, using ground-truth branch masks and RF-DETR detections

as references. The highest-F1 configuration achieved 64.74% F1, 75.16% precision and 56.86% recall, with a minimum accepted apple area of 20 px² . An alternative using a 10-px² minimum achieved a comparable F1 of 64.44%, higher recall of 61.20% and lower counting MAE of 6.68 compared with 9.00 apples/image. However, the 20-px² configuration was selected because it achieved the highest F1 and imposed a more conservative minimum candidate size, reducing susceptibility to small noise-induced detections. The complete parameter settings and development-set results are provided in Appendix Table A1.

These findings require careful interpretation because RF-DETR predictions are pseudo-ground-truth references rather than independently verified apple annotations. Higher agreement does not necessarily indicate better physical detection accuracy, particularly where RF-DETR produces false positives or misses partially occluded apples. Furthermore, a 10 px² detection is not inherently invalid, as apparent apple size depends on imaging distance and resolution. Visual inspection is therefore required to assess the plausibility of the selected configuration, while held-out evaluation is necessary to establish its generalisation performance.

Computationally, the original traditional detector required 19.48 ms for detection and 55.94 ms for complete image processing, compared with 49.28 ms and 175.98 ms for RF-DETR, respectively. The traditional approach was approximately 3.15 times faster overall, demonstrating its computational efficiency despite lower detection coverage relative to the reference.

AI-use reflection

ChatGPT was used extensively throughout the project to assist with code development, debugging, documentation and technical writing. It also supported the design of the apple-detection comparison framework, including instance-level CSV generation, one-to-one centroid matching and evaluation metrics. Codex was also used to generate documentation which was also all verified. AI was primarily used to accelerate implementation and troubleshoot technical issues, while the choice of computational approaches and interpretation of experimental results remained guided by the project objectives.

AI-generated code and recommendations were reviewed through code inspection, execution on the orchard dataset and comparison of outputs against expected behaviour. Important limitations were identified, including inconsistent branch-mask alignment between detection implementations and the risk of treating RF-DETR predictions as verified ground truth. These issues demonstrated that plausible AI-generated solutions may contain methodological assumptions requiring independent verification.

Overall, AI accelerated development but also highlighted the importance of critically evaluating generated material. This strengthened my understanding of reproducibility, validation design and the distinction between measured agreement and genuine detection accuracy. Final engineering conclusions were based on experimental results rather than AI-generated claims.

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Appendix

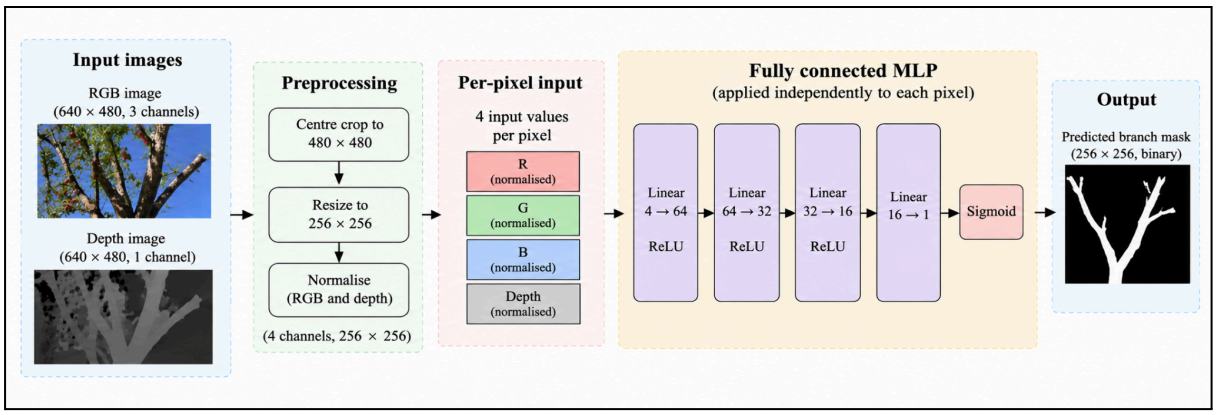


Figure A1: Left to Right MLP Baseline Architecture

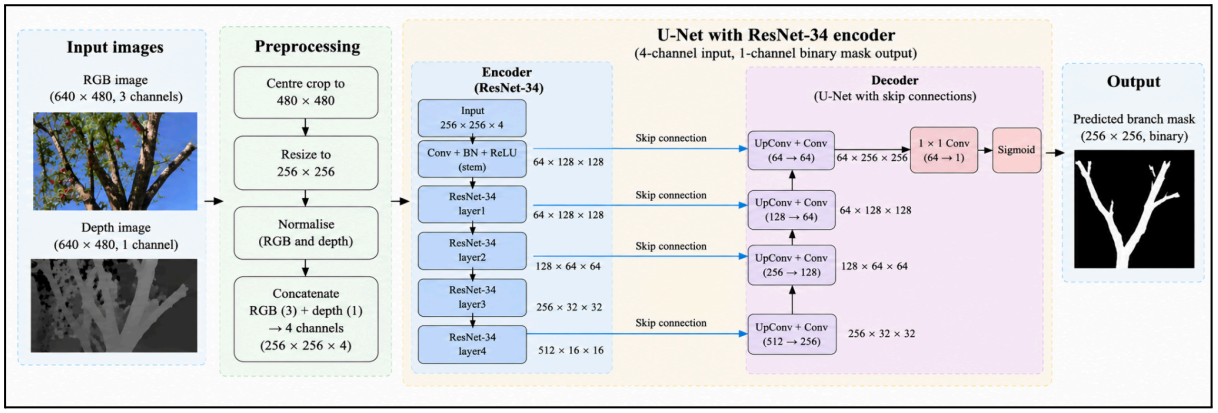


Figure A2: Left to Right Unoptimised U-Net Architecture

Table A1. Parameter configurations and development-set performance of the selected maximum-F1 detector and lower minimum area alternative (444 images; RF-DETR pseudo-ground-truth reference).

Parameter / Metric	Selected configuration	Lower min area alternative
Detection parameters		
Red intensity threshold	60	60
Red dominance ratio	1.35	1.35
Minimum apple area (px²)	**20**	10
Maximum apple area (px²)	2,000	2,000
Minimum circularity	0.10	0.10
Maximum aspect ratio	3.5	3.5
Morphological kernel size (px)	3	3
Branch dilation kernel (px)	100	100
Minimum branch overlap	0.05	0.05
Development-set results		
Mean detected apples/image	24.73	29.40
Precision (%)	**75.16**	68.04
Recall (%)	56.86	**61.20**
F1-score (%)	**64.74**	64.44
Counting MAE (apples/image)	9.00	**6.68**
Counting RMSE (apples/image)	11.05	**8.46**
Mean centroid error (px)	**5.09**	5.68
Mean bounding-box IoU (%)	**43.04**	39.91